

3 Pin Configuration

3.1 Pin Assignment - PG-VQFN-48

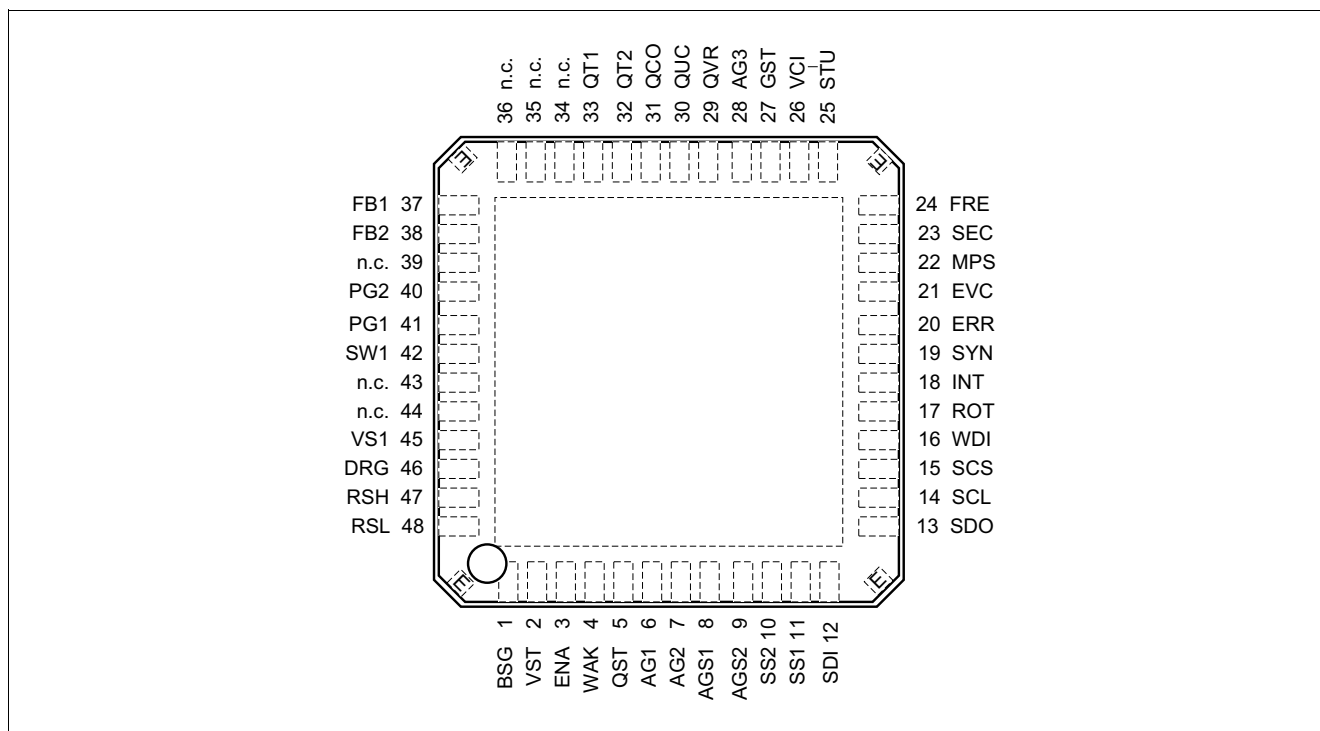


Figure 2 Pin Configuration - PG-VQFN-48

3.2 Pin Definitions and Functions - PG-VQFN-48

Pin	Symbol	Function
1	BSG	Boost driver ground: Connect this pin to ground at the low side of an external current sense resistor to decouple the driver noise from the sensitive ground. If step up pre regulator option is not used, connect to ground.
2	VST	Supply voltage standby regulator, input: Connect this input to supply (battery) voltage with reverse protection diode and capacitor between pin and ground. An EMC filter is recommended.
3	ENA	Enable Input: A positive edge signal at this pin will wake the device. In case of not used connect to ground.
4	WAK	Wake/Inhibit Input: A high level signal of defined length at this pin will wake the device. In case of not used, connect to ground.

Pin	Symbol	Function
5	QST	Output standby LDO: Connect a capacitor as close as possible to pin.
6	AG1	Analogue ground, pin 1: Connect this pin directly (low ohmic and low inductive) to ground.
7	AG2	Analogue ground, pin 2: Connect this pin directly (low ohmic and low inductive) to ground.
8	AGS1	Analogue ground, safety, pin 1: Connect this pin directly (low ohmic and low inductive) to ground. In case a safety switch is used, connect directly to the source of the NMOS used.
9	AGS2	Analogue ground, safety, pin 2: Connect this pin directly (low ohmic and low inductive) to ground. In case a safety switch is used, connect directly to the source of the NMOS used.
10	SS2	Safe state signal 2: Safe state output signal 2, sets the application into a safe state. Signal is delayed against SS1, delay can be adjusted via SPI command.
11	SS1	Safe state signal 1: Safe state output signal 1, sets the application into a safe state.
12	SDI	Serial peripheral interface, signal data input: SPI signalling port, connect to SPI port “data output” of micro processor to receive commands during SPI communication.
13	SDO	Serial peripheral interface, signal data output: SPI signalling port, connect to SPI port “data input” of micro processor to send status information during SPI communication.
14	SCL	Serial peripheral interface, signal clock: SPI signalling port, connect to SPI port “clock” of micro processor to clock the device for SPI communication.
15	SCS	Serial peripheral interface, signal chip select: SPI signalling port, connect to SPI port “chip select” of micro processor to address the device for SPI communication.
16	WDI	Watchdog input, trigger signal: Input for trigger signal, connect the “trigger signal output” of the micro processor to this pin. In case of not used, leave open (internal pull-down).
17	ROT	Reset output: Open drain structure with internal pull-up current source. A low signal at this pin indicates a reset event.
18	INT	Interrupt signal: Push-pull-stage. A low pulse at this pin indicates an interrupt, the micro processor shall read out the SPI status registers. Connect to a non maskable interrupt port (NMI) of the micro processor core supply voltage.
19	SYN	Synchronization output signal: Connect this output to the optional external switch mode post regulator synchronization input. The signal delivers the step down regulator switching frequency either in phase or shifted by 180° (selectable via SPI command). The switch mode post regulator shall synchronize to the rising edge. If not used, leave open.

Pin	Symbol	Function
20	ERR	Error signal input: Input for error signal from micro processor safety managing unit (SMU, internal failure detection of the micro processor). Connect the “error signal output” of the micro processor to this pin.
21	EVC	Enable external post regulator for core supply: Connect this pin to the enable input of the external post regulator. If not used, leave open.
22	MPS	Microcontroller programming support pin: Pull down this pin to ground for operation. Optionally, this pin can be used for microcontroller debugging and programming purposes. For details please refer to Chapter 11.7 .
23	SEC	Configuration pin for external post regulator for core supply: Connect this pin to ground if the option external post regulator is not used. If the option external post regulator is used, leave open.
24	FRE	Frequency adjustment pin: Connect pin to ground for low frequency range or leave open for high frequency range.
25	STU	Configuration pin for step up converter: Connect this pin to ground if the option step up pre regulator is not used. If the option step up pre regulator is used, leave open.
26	VCI	Input for optional external post regulator output voltage (core supply): Connect an external resistor divider to adjust the over and under voltage thresholds of reset output signal ROT. If the option external post regulator is not used, leave open.
27	GST	Gate stress pin: Not for customer use. Connect this pin directly (low ohmic and low inductive) to ground.
28	AG3	Analogue ground, pin 3: Connect this pin directly (low ohmic and low inductive) to ground.
29	QVR	Output voltage reference: Connect a capacitor as close as possible to pin.
30	QUC	Output LDO_uC supply (micro processor supply): Connect a capacitor as close as possible to pin.
31	QCO	Output LDO_communication supply: Connect a capacitor as close as possible to pin.
32	QT2	Output tracker 2: Connect a capacitor as close as possible to pin.
33	QT1	Output tracker 1: Connect a capacitor as close as possible to pin.
34	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.

Pin	Symbol	Function
35	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
36	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
37	FB1	Step down pre regulator feedback input plus input for linear post regulators and trackers, pin 1: Connect the capacitor of the step down pre regulator output filter with low ohmic and low inductive connection straight to this pin. Always connect in parallel with pin FB2.
38	FB2	Step down pre regulator feedback input plus input for linear post regulators and trackers, pin 2: Connect the capacitor of the step down pre regulator output filter with low ohmic and low inductive connection straight to this pin. Always connect in parallel with pin FB1.
39	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
40	PG2	Step down pre regulator power ground, pin 2: Connect this pin straight (low ohmic and low inductive) to ground and pre regulator output capacitor minus. Always connect in parallel with pin PG1.
41	PG1	Step down pre regulator power ground, pin 1: Connect this pin straight (low ohmic and low inductive) to ground and pre regulator output capacitor minus. Always connect in parallel with pin PG2.
42	SW1	Step down pre regulator power stage output: Connect this pin straight (low ohmic and low inductive) to the pre regulator output filter.
43	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
44	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
45	VS1	Supply voltage step down pre regulator input: Connect this input to the output of the step up pre regulator. If step up pre regulator option is not used, connect to supply (battery) voltage with reverse protection diode and capacitor between pin and ground. An EMC filter is recommended.

Pin	Symbol	Function
46	DRG	Driver output for external step up regulator power stage, connect to gate: Gate of low side switch of step up pre regulator: Connect to the gate of an external N-channel mosfet, line to be straight and as short as possible. If step up pre regulator option is not used, leave open.
47	RSH	Sense resistor for external step up regulator power stage, high side: Connect this pin to the high side of an external current sense resistor to determine the maximum current threshold through the external N-channel mosfet. If step up pre regulator option is not used, connect to ground.
48	RSL	Sense resistor for external step up regulator power stage, low side: Connect this pin to the low side of an external current sense resistor to determine the maximum current threshold through the external N-channel mosfet. If step up pre regulator option is not used, connect to ground.
	EP1	Edge pin no 1: Keep area below this pin free of ground or other signals, do not solder this pin to ground or any other signal. This pin must be kept free of soldering.
	EP2	Edge pin no 2: Keep area below this pin free of ground or other signals, do not solder this pin to ground or any other signal. This pin must be kept free of soldering.
	EP3	Edge pin no 3: Keep area below this pin free of ground or other signals, do not solder this pin to ground or any other signal. This pin must be kept free of soldering.
	EP4	Edge pin no 4: Keep area below this pin free of ground or other signals, do not solder this pin to ground or any other signal. This pin must be kept free of soldering.
Cooling Tab	GND	Cooling Tab. Connect externally to GND and heat sink area.

3.3 Pin Assignment - PG-LQFP-64

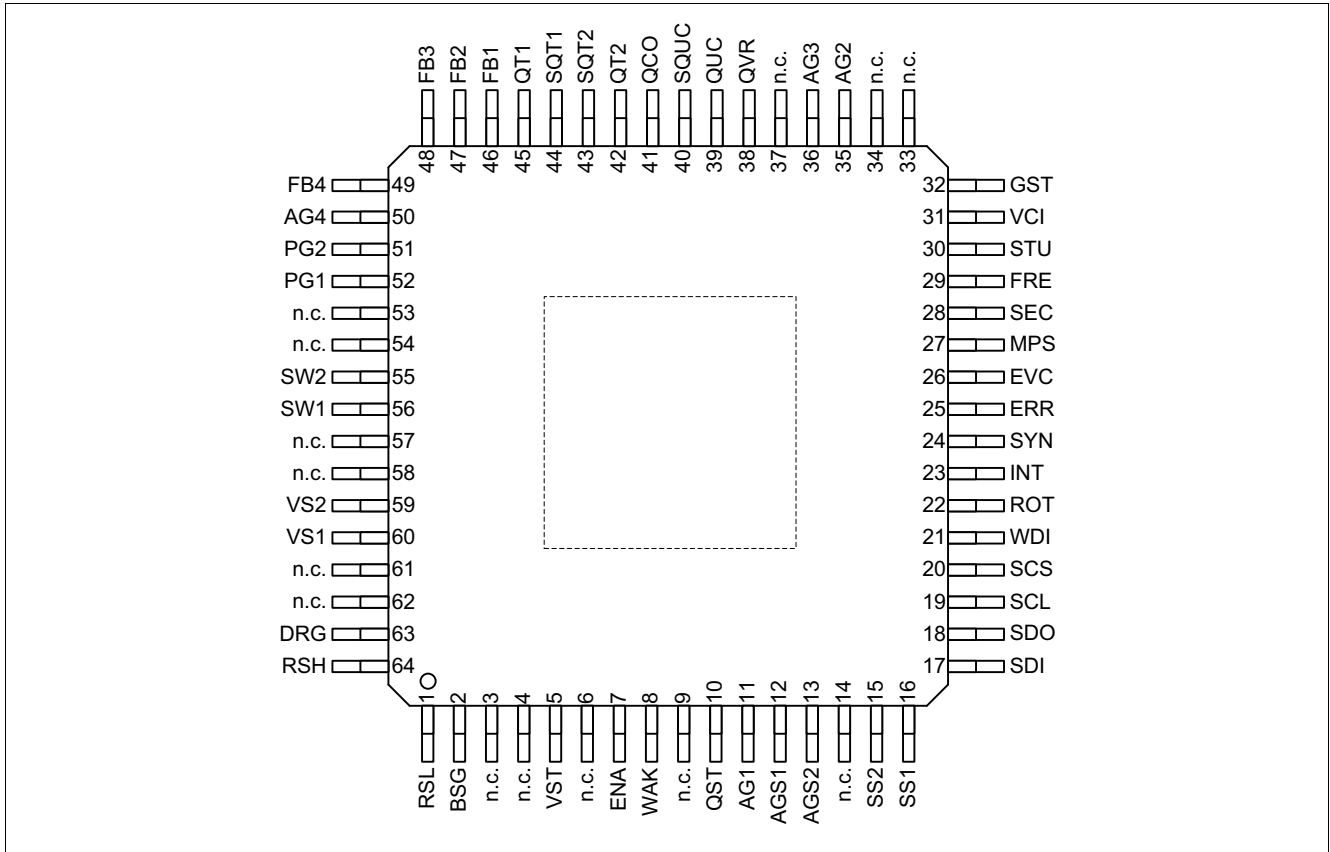


Figure 3 Pin Configuration - PG-LQFP-64

3.4 Pin Definitions and Functions - PG-LQFP-64

Pin	Symbol	Function
1	RSL	Sense resistor for external step up regulator power stage, low side: Connect this pin to the low side of an external current sense resistor to determine the maximum current threshold through the external N-channel mosfet. If step up pre regulator option is not used, connect to ground.
2	BSG	Boost driver ground: Connect this pin to ground at the low side of an external current sense resistor to decouple the driver noise from the sensitive ground. If step up pre regulator option is not used, connect to ground.
3	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.

Pin	Symbol	Function
4	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
5	VST	Supply voltage standby regulator, input: Connect this input to supply (battery) voltage with reverse protection diode and capacitor between pin and ground. An EMC filter is recommended.
6	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
7	ENA	Enable Input: A positive edge signal at this pin will wake the device. In case of not used, connect to ground.
8	WAK	Wake/Inhibit Input: A high level signal of defined length at this pin will wake the device. In case of not used, connect to ground.
9	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
10	QST	Output standby LDO: Connect a capacitor as close as possible to pin.
11	AG1	Analogue ground, pin 1: Connect this pin directly (low ohmic and low inductive) to ground.
12	AGS1	Analogue ground, safety, pin 1: Connect this pin directly (low ohmic and low inductive) to ground. In case a safety switch is used, connect directly to the source of the NMOS used.
13	AGS2	Analogue ground, safety, pin 2: Connect this pin directly (low ohmic and low inductive) to ground. In case a safety switch is used, connect directly to the source of the NMOS used.
14	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
15	SS2	Safe state signal 2: Safe state output signal 2, sets the application into a safe state. Signal is delayed against SS1, delay can be adjusted via SPI command.
16	SS1	Safe state signal 1: Safe state output signal 1, sets the application into a safe state.
17	SDI	Serial peripheral interface, signal data input: SPI signalling port, connect to SPI port “data output” of micro processor to receive commands during SPI communication.
18	SDO	Serial peripheral interface, signal data output: SPI signalling port, connect to SPI port “data input” of micro processor to send status information during SPI communication.

Pin	Symbol	Function
19	SCL	Serial peripheral interface, signal clock: SPI signalling port, connect to SPI port “clock” of micro processor to clock the device for SPI communication.
20	SCS	Serial peripheral interface, signal chip select: SPI signalling port, connect to SPI port “chip select” of micro processor to address the device for SPI communication.
21	WDI	Watchdog input, trigger signal: Input for trigger signal, connect the “trigger signal output” of the micro processor to this pin. In case of not used, leave open (internal pull-down).
22	ROT	Reset output: Open drain structure with internal pull-up current source. A low signal at this pin indicates a reset event.
23	INT	Interrupt signal: Push-pull-stage. A low pulse at this pin indicates an interrupt, the micro processor shall read out the SPI status registers. Connect to a non maskable interrupt port (NMI) of the micro processor core supply voltage.
24	SYN	Synchronization output signal: Connect this output to the optional external switch mode post regulator synchronization input. The signal delivers the step down regulator switching frequency either in phase or shifted by 180° (selectable via SPI command). The switch mode post regulator shall synchronize to the rising edge. If not used, leave open.
25	ERR	Error signal input: Input for error signal from micro processor safety managing unit (SMU, internal failure detection of the micro processor). Connect the “error signal output” of the micro processor to this pin.
26	EVC	Enable external post regulator for core supply: Connect this pin to the enable input of the external post regulator. If not used, leave open.
27	MPS	Microcontroller programming support pin: Pull down this pin to ground for operation. Optionally, this pin can be used for microcontroller debugging and programming purposes. For details please refer to Chapter 11.7 .
28	SEC	Configuration pin for external post regulator for core supply: Connect this pin to ground if the option external post regulator is not used. If the option external post regulator is used, leave open.
29	FRE	Frequency adjustment pin: Connect pin to ground for low frequency range or leave open for high frequency range.
30	STU	Configuration pin for step up converter: Connect this pin to ground if the option step up pre regulator is not used. If the option step up pre regulator is used, leave open.
31	VCI	Input for optional external post regulator output voltage (core supply): Connect an external resistor divider to adjust the over and under voltage thresholds of reset output signal ROT. If the option external post regulator is not used, leave open.

Pin	Symbol	Function
32	GST	Gate stress pin: Not for customer use. Connect this pin directly (low ohmic and low inductive) to ground.
33	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
34	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
35	AG2	Analogue ground, pin 2: Connect this pin directly (low ohmic and low inductive) to ground.
36	AG3	Analogue ground, pin 3: Connect this pin directly (low ohmic and low inductive) to ground.
37	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
38	QVR	Output voltage reference: Connect a capacitor as close as possible to pin.
39	QUC	Output LDO_uC supply (micro processor supply): Connect a capacitor as close as possible to pin.
40	SQUC	Sense connection for LDO_uC supply (micro processor supply): Connect to QUC/LDO_uC externally.
41	QCO	Output LDO_communication supply: Connect a capacitor as close as possible to pin.
42	QT2	Output tracker 2: Connect a capacitor as close as possible to pin.
43	SQT2	Sense connection tracker 2: Connect to QT2/tracker 2 externally.
44	SQT1	Sense connection tracker 1: Connect to QT1/tracker 1 externally.
45	QT1	Output tracker 1: Connect a capacitor as close as possible to pin.
46	FB1	Step down pre regulator feedback input plus input for linear post regulators and trackers, pin 1: Connect the capacitor of the step down pre regulator output filter with low ohmic and low inductive connection straight to this pin. Always connect in parallel with pin FB1 - FB4.
47	FB2	Step down pre regulator feedback input plus input for linear post regulators and trackers, pin 2: Connect the capacitor of the step down pre regulator output filter with low ohmic and low inductive connection straight to this pin. Always connect in parallel with pin FB1 - FB4.

Pin	Symbol	Function
48	FB3	Step down pre regulator feedback input plus input for linear post regulators and trackers, pin 3: Connect the capacitor of the step down pre regulator output filter with low ohmic and low inductive connection straight to this pin. Always connect in parallel with pin FB1 - FB4.
49	FB4	Step down pre regulator feedback input plus input for linear post regulators and trackers, pin 4: Connect the capacitor of the step down pre regulator output filter with low ohmic and low inductive connection straight to this pin. Always connect in parallel with pin FB1 - FB4.
50	AG4	Analogue ground, pin 4: Connect this pin directly (low ohmic and low inductive) to ground.
51	PG2	Step down pre regulator power ground, pin 2: Connect this pin straight (low ohmic and low inductive) to ground and pre regulator output capacitor minus. Always connect in parallel with pin PG1.
52	PG1	Step down pre regulator power ground, pin 1: Connect this pin straight (low ohmic and low inductive) to ground and pre regulator output capacitor minus. Always connect in parallel with pin PG2.
53	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
54	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
55	SW2	Step down pre regulator power stage output, pin 2: Connect this pin straight (low ohmic and low inductive) to the pre regulator output filter. Always connect in parallel with pin SW1.
56	SW1	Step down pre regulator power stage output, pin 1: Connect this pin straight (low ohmic and low inductive) to the pre regulator output filter. Always connect in parallel with pin SW2.
57	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
58	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
59	VS2	Supply voltage step down pre regulator, pin 2, input: Connect this input in parallel with VS1 to the output of the step up pre regulator. If step up pre regulator option is not used, connect in parallel with VS1 to supply (battery) voltage with reverse protection diode and capacitor between pin and ground. An EMC filter is recommended.

Pin	Symbol	Function
60	VS1	Supply voltage step down pre regulator, pin 1, input: Connect this input in parallel with VS2 to the output of the step up pre regulator. If step up pre regulator option is not used, connect in parallel with VS2 to supply (battery) voltage with reverse protection diode and capacitor between pin and ground. An EMC filter is recommended.
61	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
62	N.C.	Internally not connected: This pin is electrically not connected internally and can be kept open/floating, connected to GND or any other signal. Consider neighboring signals for potential failures.
63	DRG	Driver output for external step up regulator power stage, connect to gate: Gate of low side switch of step up pre regulator: Connect to the gate of an external N-channel mosfet, line to be straight and as short as possible. If step up pre regulator option is not used, leave open.
64	RSH	Sense resistor for external step up regulator power stage, high side: Connect this pin to the high side of an external current sense resistor to determine the maximum current threshold through the external N-channel mosfet. If step up pre regulator option is not used, connect to ground.
Cooling Tab	GND	Cooling Tab. Connect externally to GND and heat sink area.